

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 1 – EXPERIMENTS

|                  |                                                                                                          |               |                                   |
|------------------|----------------------------------------------------------------------------------------------------------|---------------|-----------------------------------|
| Course Code:     | CSA06                                                                                                    | Course Title: | Design and Analysis of Algorithms |
| CO Assessed:     | <b>CO3: Articulation</b><br>(BL4, BL5)                                                                   | Assessment:   | Assessment Tool 1 – Experiments   |
| Weightage:       | 40% of CIA                                                                                               | Total Marks:  | 100                               |
| CO3 Statement:   | Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication |               |                                   |
| Student Name:    | CH.BHAVYA SRI                                                                                            | Reg. No.:     | 192411249                         |
| Section / Batch: | C                                                                                                        | Date:         | 1-08-2026                         |

### Instructions to Students

- Answer the following question. Total Marks: 100.
- Write the algorithm and execute the code for the given experiment.
- Follow a well-structured, systematic approach to design and implement an optimal solution.
- Provide insightful analysis with accurate validation, supported by clear documentation including diagrams, code, and results.

| Question Type | Focus Area                                                                                                          | Questions |
|---------------|---------------------------------------------------------------------------------------------------------------------|-----------|
| Experiments   | Design and Code for Divide and Conquer Algorithms: Merge Sort, Quick Sort, Binary Search, and Matrix Multiplication | Q1        |

### SECTION A – Experiments

Analyze and compare the performance of Binary Search and Linear Search by implementing both algorithms on datasets of varying sizes. Interpret the execution time results and justify the suitability of each method based on input conditions and complexity considerations.

### Code of Conduct

I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: \_\_\_\_\_

## RUBRICS FOR CALCULATION

| Experiments               |           |                                                                                               |                                                             |                                                     |                                                   |
|---------------------------|-----------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|---------------------------------------------------|
| Criteria                  | Weightage | Level 4 – Exemplary                                                                           | Level 3 – Proficient                                        | Level 2 – Developing                                | Level 1 – Beginning                               |
| Understanding of Concepts | 25%       | Demonstrates thorough understanding and effectively integrates multiple concepts/technologies | Good understanding with proper integration of most concepts | Basic understanding; limited or partial integration | Poor understanding ; unable to integrate concepts |
| Experimental Design       | 30%       | Well-planned, systematic implementation with optimal solution                                 | Correct implementation with minor issues                    | Basic implementation with errors or inefficiencies  | Incorrect or incomplete implementation            |
| Use of Tools              | 20%       | Effective and advanced use of tools/technologies with proper configuration                    | Appropriate use of tools with correct execution             | Limited or inconsistent use of tools                | Incorrect or minimal use of tools                 |
| Analysis of Results       | 15%       | Insightful analysis with accurate interpretation and validation of results                    | Good analysis with reasonable interpretation                | Basic analysis with limited insights                | Poor or incorrect analysis of results             |
| Documentation             | 10%       | Clear, well-structured documentation with diagrams, code, and results                         | Organized documentation with necessary details              | Basic documentation with missing details            | Poor or incomplete documentation                  |

### Marks Evaluation:

| Criteria                  | Wt. | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---------------------------|-----|---------------------|----------------------|----------------------|---------------------|
| Understanding of Concepts | 25% |                     |                      |                      |                     |
| Experimental Design       | 30% |                     |                      |                      |                     |
| Use of Tools              | 20% |                     |                      |                      |                     |
| Analysis of Results       | 15% |                     |                      |                      |                     |
| Documentation             | 10% |                     |                      |                      |                     |
| <b>Total Marks (100):</b> |     |                     |                      |                      |                     |

| Faculty In-charge | Course Coordinator | Head of Department |
|-------------------|--------------------|--------------------|
| Name:             | Name:              | Name:              |
| Signature: _____  | Signature: _____   | Signature: _____   |
| Date: _____       | Date: _____        | Date: _____        |

## QUESTIONS

Q1

Binary Search vs Linear Search  
100 marks

1/1 answered

## Q1 Binary Search vs Linear Search

100 marks

Analyze and compare the performance of Binary Search and Linear Search by implementing both algorithms on datasets of varying sizes. Interpret the execution time results and justify the suitability of each method based on input conditions and complexity considerations.

## Your Code

```
1 #include<stdio.h>
2 int main() {
3     int a[20], n, x, i, low, high, mid;
4     printf("Enter n:");
5     scanf("%d", &n);
6     printf("Entered stored element:\n");
7     for(i=0; i<n; i++)
8         scanf("%d", &a[i]);
9     printf("Enter element");
10    scanf("%d", &x);
11    for(i=0; i<n; i++)
12        if(a[i]==x)
13            break;
14    if(i<n)
15        printf("linear search found in %d\n", i);
16    else
```

## Input

```
5
10 20 30 40 50
40
```

## Output

```
Enter n:Entered stored element:
Enter elementlinear search found in
3
binary search: found at 3
```

Activate Windows  
Go to Settings to activate Windows.